RAJALAKSHMI ENGINEERING COLLEGE

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C2

Started on	Thursday, 17 September 2026, 8:30 AM
State	Finished
Completed on	Thursday, 17 September 2026, 8:30 AM
Time taken	50 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that A[j] - A[i] = k, i != j.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as 5 - 1 = 4

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```
#include <stdio.h>

#define MAX_N 100005

int A[MAX_N];

int main() {
    int n;
    // Read the total number of elements (handles line spaces or newlines automatically)
    if (scanf("%d", &n) != 1) return 0;

    // Read the n elements of the array
    for (int idx = 0; idx < n; idx++) {
        if (scanf("%d", &A[idx]) != 1) return 0;
    }

    int k;
    // Read the target non-negative integer difference k
    if (scanf("%d", &k) != 1) return 0;

    int i = 0;
    int j = 1;
    int found = 0;

    // Two-pointer linear traversal
    while (i < n && j < n) {
        int diff = A[j] - A[i];

        if (i != j && diff == k) {
            found = 1;
            break;
        } else if (diff < k) {
```

```
            j++;
        } else {
            i++;
        }

        // Ensure the two pointers never overlap on the exact same index
        if (i == j) {
            j++;
        }
    }

    // Output 1 if a pair exists, otherwise 0
    printf("%d\n", found);

    return 0;
}
```

	Input	Expected	Got	
✔	3 1 3 5 4	1	1	✔
✔	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✔
✔	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✔
✔	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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